

TURYN SEQUENCES – CONSOLIDATED THEORY SUMMARY (PLAIN TEXT)

1. DEFINITIONS

A Turyn sequence is a quadruple of binary sequences (A, B, C, D), each using ± 1 entries, satisfying a set of autocorrelation cancellation conditions. They are connected to Golay pairs, complementary sequences, periodic autocorrelation, and supplementary difference sets (SDS).

2. AUTOCORRELATION

For a sequence X of length n:

Nonperiodic autocorrelation: $NX(s) = \sum_i x_i \cdot x_{i+s}$ (with out-of-range = 0)

Periodic autocorrelation: $PX(s) = \sum_i x_i \cdot x_{(i+s \bmod n)}$

Complementary sequences satisfy $NX(s) + NY(s) = 0$ for all $s \neq 0$.

Turyn sequences generalize these by forming a 4-tuple that satisfies a specific weighted sum of autocorrelations equaling 0.

3. TURYN SEQUENCE CONDITIONS

Let A, B, C, D be binary sequences of lengths n_A, n_B, n_C, n_D .

A valid Turyn quadruple satisfies:

$TA(s) + TB(s) + 2 \cdot TC(s) + 2 \cdot TD(s) = 0$ for all nonzero shifts s

Where $T^*(s)$ is the nonperiodic autocorrelation of each sequence.

4. CANONICAL FORMS

To avoid redundant equivalent sequences, impose:

- First element of each sequence = +1
- A is lexicographically smallest among equivalent sequences.
- Symmetry reductions: reversal, negation, ($X \rightarrow -X$), and alternating sign flips.

These canonical reductions reduce search space and avoid duplicates.

5. EQUIVALENCE OPERATIONS

Allowed transformations that preserve Turyn properties:

- Negation of any sequence
- Reversal of sequences
- Alternating ($x_i \rightarrow (-1)^i x_i$)

Used to canonicalize prefixes and prune illegal branches.

6. PREFIX CONSTRAINTS

When generating prefixes:

- Partial autocorrelation constraints must not already violate Turyn cancellation.
- Each partial prefix must not violate symmetry rules.
- If any shift s violates $TA(s)+TB(s)+2TC(s)+2TD(s)=0$ in partial form (overlapping part), the prefix is invalid.

7. FORBIDDEN PATTERNS

From complementary and Golay theory:

- Long runs of +1 or -1 at early positions are typically forbidden because they cause autocorrelation spikes.
- Prefixes where two sequences start with identical long runs often create violations.
- Certain combinations of (A[i], B[i], C[i], D[i]) cannot simultaneously be +1 if they force autocorrelation coefficients that cannot cancel later.

8. SDS (SUPPLEMENTARY DIFFERENCE SETS) LINK

Pairs of binary sequences correspond to subsets $X \subseteq \mathbb{Z}_n$ where -1 entries represent set membership.

Complementary and periodic Golay conditions correspond to difference-set constraints.

These impose restrictions on the distribution of ± 1 values across the sequences.

9. GOLAY SEQUENCE THEORY (RELEVANT TO TURYN)

Golay pairs (A,B) satisfy:

$$NA(s) + NB(s) = 0$$

Turyn quadruples generalize this to a weighted sum over 4 sequences.

Properties:

- Length constraints impose parity restrictions.
- Sequences must balance numbers of +1 and -1 to meet autocorrelation cancellation.
- Specific structural constraints (from Golay pair literature) forbid certain early patterns.

10. THEORY-GUIDED PREFIX GENERATION RULES

When generating candidate prefixes:

- Maintain canonical form (first element +1).
- Apply symmetry pruning immediately.
- For each shift s up to prefix length, compute partial autocorrelation for A, B, C, D and reject prefix if:

$$TA_p(s) + TB_p(s) + 2 \cdot TC_p(s) + 2 \cdot TD_p(s) \neq 0 \text{ (over the overlapping prefix).}$$

- Ensure the marginal autocorrelation contributions remain correctable by future positions.
- Avoid symmetric duplicates via lexicographic checks.

11. PRACTICAL PREFIX RULES FOR N=5 AND N=7

Empirical constraints observed in known small Turyn and Golay sequences:

- Early sequences rarely place identical triple (+1,+1,+1) across A,B,C at same index.
- Balanced transitions (+1 $\rightarrow$ -1) should appear early for valid autocorrelation structure.
- For N=7, sequences typically show alternating or near-alternating patterns in C and D.

12. APPLICATION TO SAT-CHECKABLE PREFIXES

SAT encoding uses CNF clauses to enforce:

- Variable constraints $x[i] \in \{\pm 1\}$
- Autocorrelation cancellation constraints
- Canonical symmetry constraints
- Prefix assignments as assumptions

NECESSARY PREFIX PROPERTIES FOR SAT-VIABILITY:

- No autocorrelation violation at any shift within the prefix.
- No violation of canonical form.
- Prefix must not force incompatible parity or SDS constraints.

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